



BLM3620 Digital Signal Processing

Dr. Ali Can KARACA

ackaraca@yildiz.edu.tr

Yıldız Technical University – Computer Engineering

Lecture #4 – Sampling and Aliasing

- Sampling
- Principal of Aliasing
- Spectrum of a Discrete-Time Signal
- Over-Sampling & Under-Sampling
- Stroboscopic effect

Important Materials:

- James H. McClellan, R. W. Schafer, M. A. Yoder, *DSP First Second Edition*, Pearson, 2015.
- Lizhe Tan, Jean Jiang, *Digital Signal Processing: Fundamentals and Applications*, Third Edition, Academic Press, 2019.

Auxiliary Materials:

- Prof. Sarp Ertürk, *Sayısal İşaret İşleme*, Birsen Yayınevi.
- Prof. Nizamettin Aydın, DSP Lecture Notes.
- J. G. Proakis, D. K. Manolakis, *Digital Signal Processing Fourth Edition*, Pearson, 2014.
- J. K. Perin, *Digital Signal Processing, Lecture Notes*, Stanford University, 2018.

Syllabus



Week	Lectures
1	Introduction to DSP and MATLAB
2	Sinuzoids and Complex Exponentials
3	Spectrum Representation
4	Sampling and Aliasing
5	Discrete Time Signal Properties and Convolution
6	Convolution and FIR Filters
7	Frequency Response of FIR Filters
8	Midterm Exam
9	Discrete Time Fourier Transform and Properties
10	Discrete Fourier Transform and Properties
11	Fast Fourier Transform and Windowing
12	z- Transforms
13	FIR Filter Design and Applications
14	IIR Filter Design and Applications
15	Final Exam

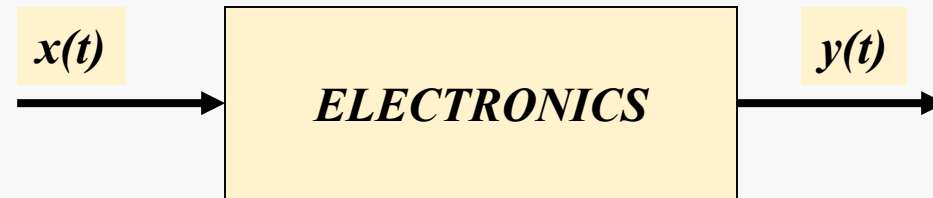
For more details -> Bologna page: <http://www.bologna.yildiz.edu.tr/index.php?r=course/view&id=5730&aid=3>

Remember: Analog & Digital Systems



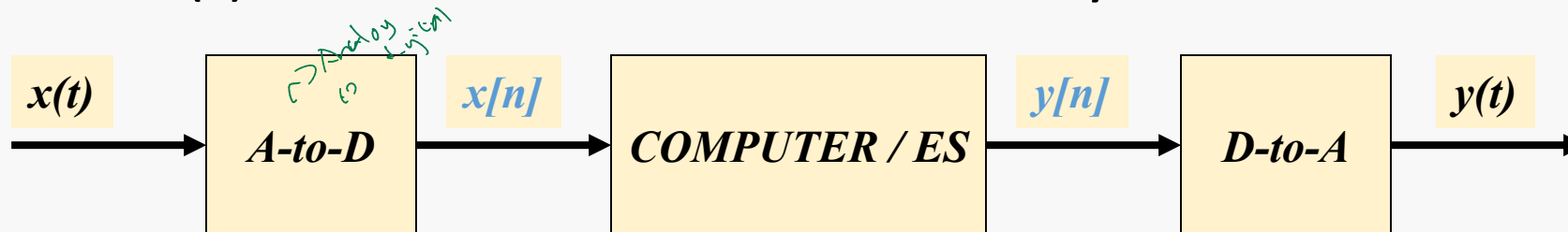
- **ANALOG/ELECTRONIC:**

- Circuits: resistors, capacitors, op-amps



- **DIGITAL/MICROPROCESSOR**

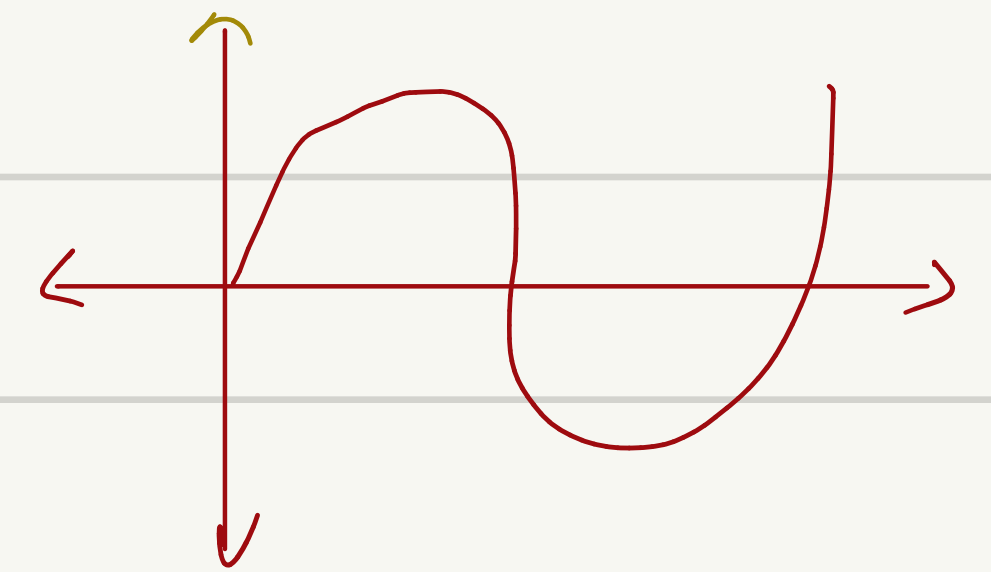
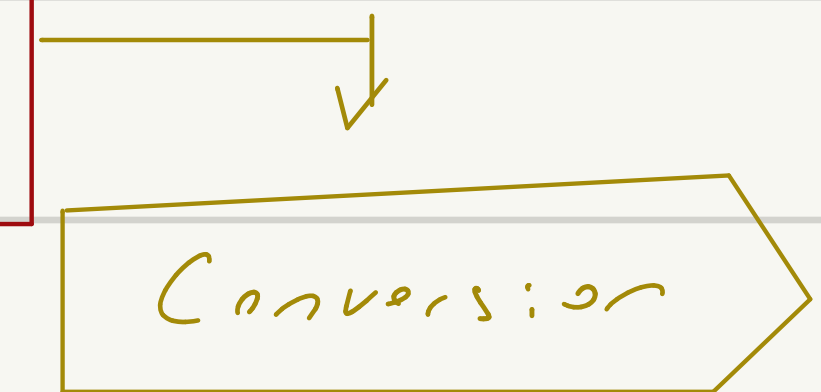
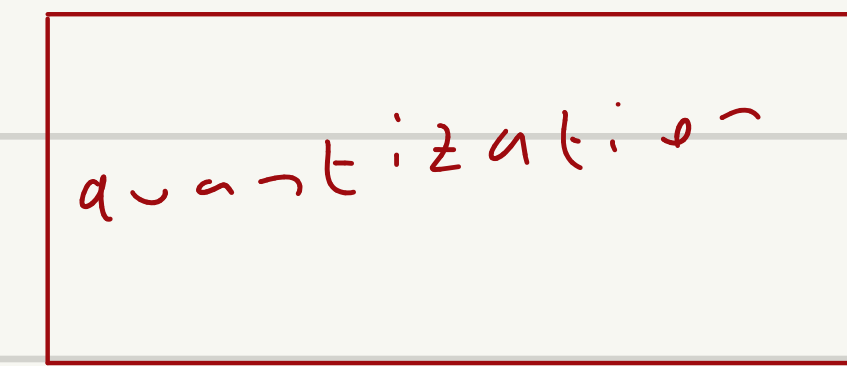
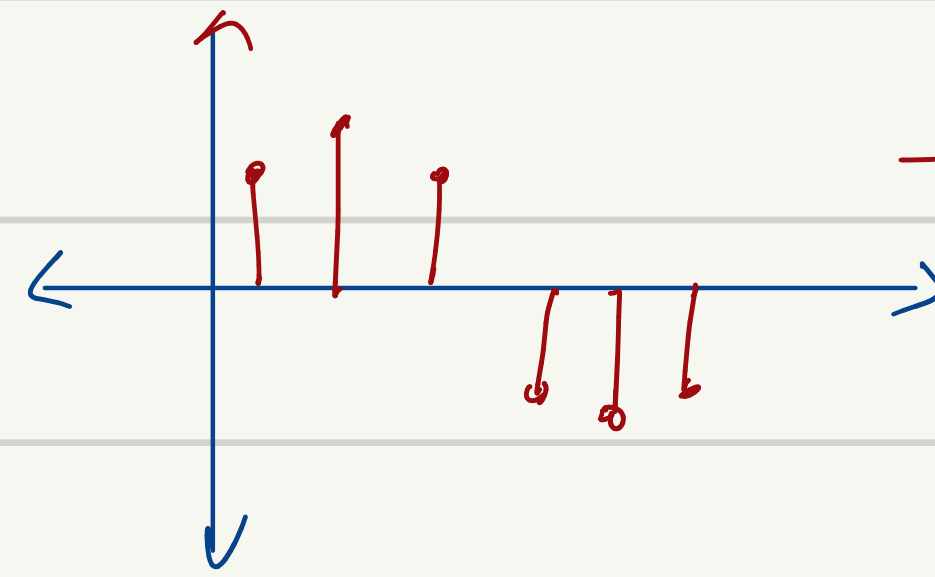
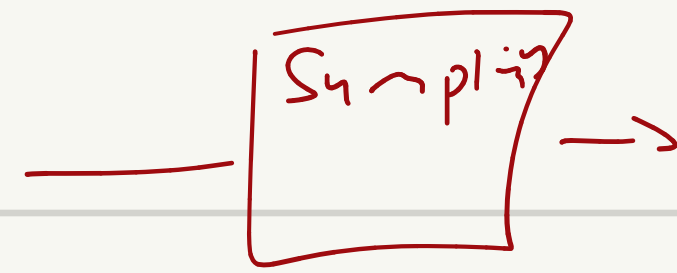
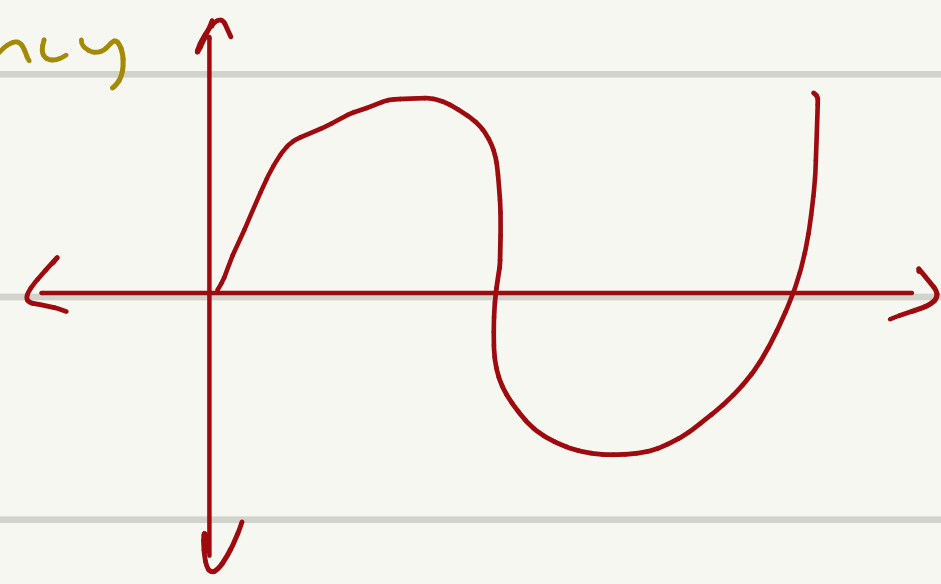
- Convert $x(t)$ to **numbers** stored in memory



Steps of Analog to digital conversion

1-) Sampling frequency

2-) Quantization level



Sampling of Analog Signals



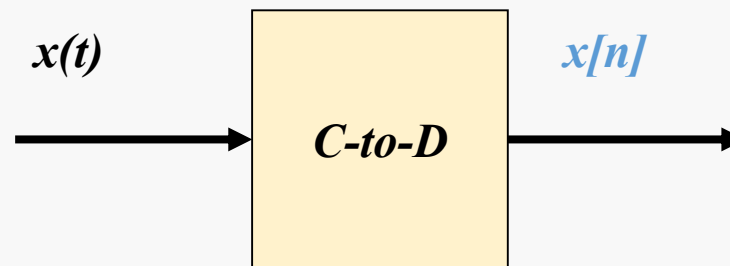
- SAMPLING PROCESS

- Convert $x(t)$ to **numbers** $x[n]$
- “n” is an integer index; $x[n]$ is a sequence of values
- Think of “n” as the storage address in memory

$$t = n \cdot T_s$$

- UNIFORM SAMPLING at $t = nT_s$

- IDEAL: $x[n] = x(nT_s)$



Sampling of Analog Signals



- SAMPLING RATE (f_s)

- $f_s = 1/T_s$

- NUMBER of SAMPLES PER SECOND

- $T_s = 125 \text{ microsec} \rightarrow f_s = 8000 \text{ samples/sec}$

- UNITS of f_s ARE HERTZ: 8000 Hz

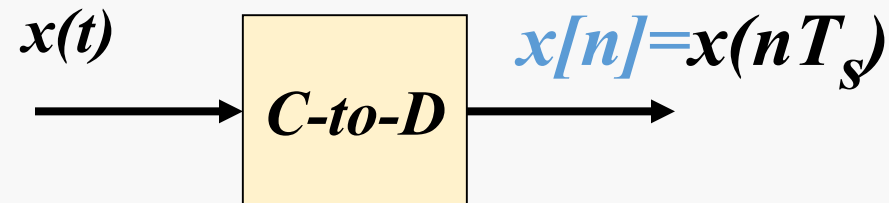
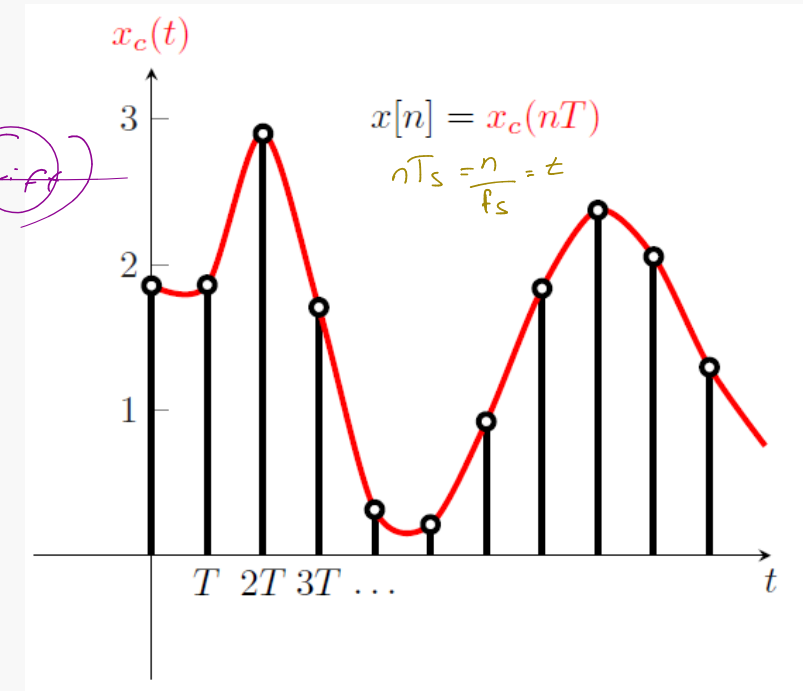
- UNIFORM SAMPLING at $t = nT_s = n/f_s$

- IDEAL: $x[n] = x(nT_s) = x(n/f_s)$

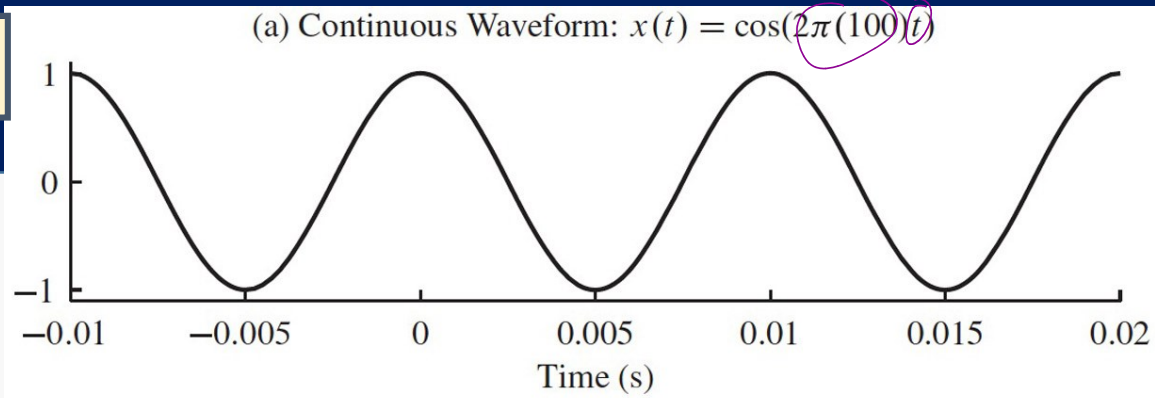
Handwritten notes:

$$\omega = 2\pi f$$
$$n \cdot T_s = t$$
$$n = t \cdot f_s$$

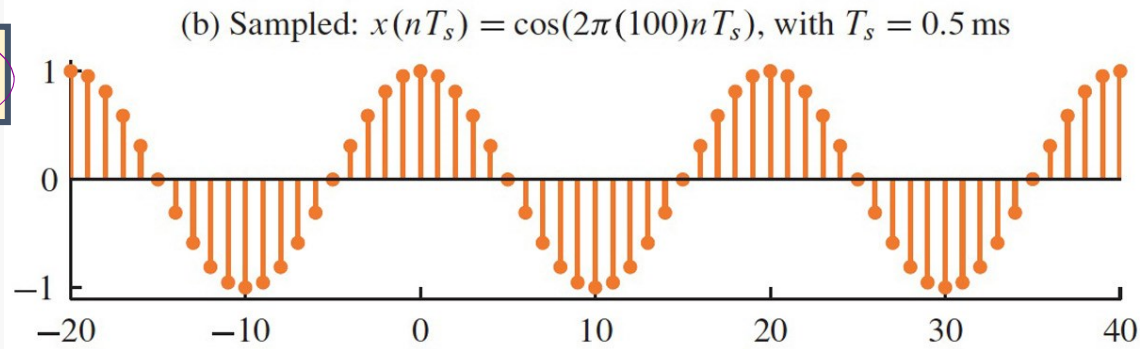
(circled in purple)



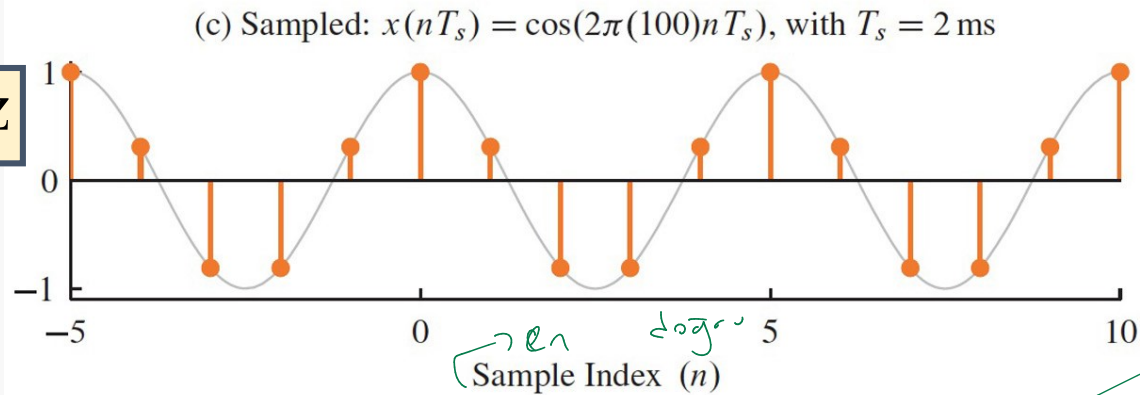
$$f = 100\text{Hz}$$



$$f_s = 2\text{kHz}$$



$$f_s = 500\text{Hz}$$



Handwritten notes in purple ink:

- $n \cdot f_s$
- $\frac{n}{f_s}$
- $\frac{1}{2000} \cdot 127 \cdot 1000$
- $0.5 \cdot \pi \cdot n$
- $0.25 \cdot 2 \cdot \pi \cdot n$
- $\frac{1}{500} \cdot 27 \cdot 1000$
- $0.2 \cdot 2 \cdot \pi \cdot n$

Which one provides the most accurate representation of $x(t)$?

Sampling Theorem



- HOW OFTEN DO WE NEED TO SAMPLE?
 - DEPENDS on FREQUENCY of SINUSOID
 - ANSWERED by SHANNON/NYQUIST Theorem
 - ALSO DEPENDS on "RECONSTRUCTION"

$$f_s = \frac{1}{T_s} \rightarrow 2f_{\max}$$

Shannon Sampling Theorem

A continuous-time signal $x(t)$ with frequencies no higher than $f_{\max}$ can be reconstructed exactly from its samples $x[n] = x(nT_s)$, if the samples are taken at a rate $f_s = 1/T_s$ that is greater than $2f_{\max}$.

- “Nyquist Rate” Sampling

- $f_s > \underline{\text{TWICE}}$ the HIGHEST Frequency in $x(t)$
- “Sampling above the Nyquist rate”

$$X(t) = \cos(2\pi 100t) \rightarrow f_s \geq 100 \times 2$$

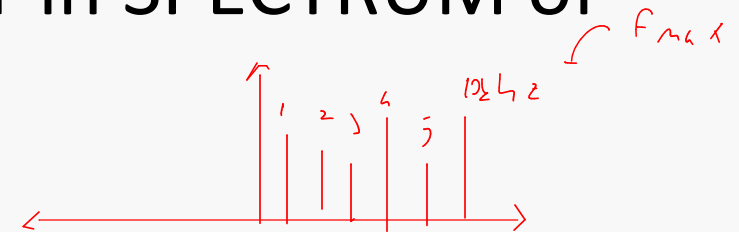
$$X(t) = 2\cos(10\pi t + \frac{\pi}{2}) + \cos(20\pi t + \pi)$$

$f = 5$ $f = 10$

$$f_s \geq 2 \times 10 \text{ Hz}$$

- **BANDLIMITED SIGNALS**

- DEF: HIGHEST FREQUENCY COMPONENT in SPECTRUM of $x(t)$ is finite
- NON-BANDLIMITED EXAMPLE
 - TRIANGLE WAVE is **NOT** BANDLIMITED



Discrete-Time Sinusoid



- Change $x(t)$ into $x[n]$ **DERIVATION**

$$x(t) = A \cos(\omega t + \varphi)$$

$$x[n] = x(nT_s) = A \cos(\omega nT_s + \varphi)$$

$$x[n] = A \cos((\omega T_s)n + \varphi)$$

$$x[n] = A \cos(\hat{\omega}n + \varphi)$$

$$\hat{\omega} = \omega T_s = \frac{\omega}{f_s}$$

DEFINE DIGITAL FREQUENCY

$$\hat{\omega} = \frac{\omega}{f_s}$$

$$\frac{2\pi f}{f_s} = \hat{\omega}$$

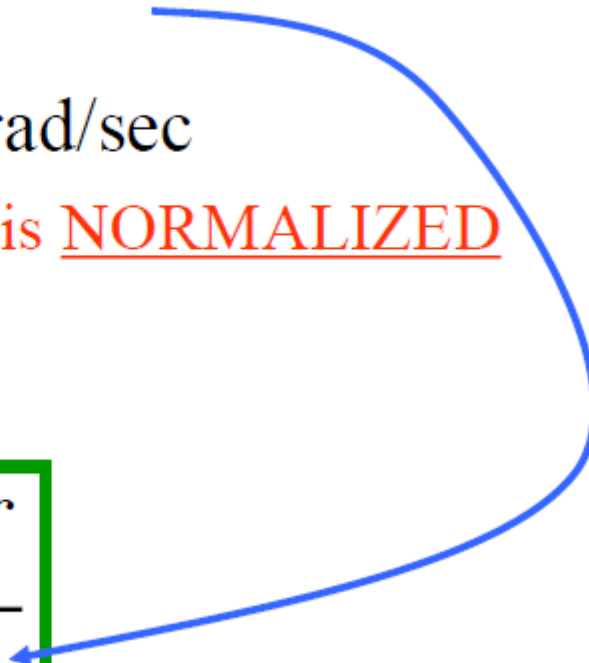
$$\frac{2\pi f}{f_s}$$

$$\hat{\omega} = \frac{2\pi f}{f_s} \text{ Hz} \rightarrow \text{radian}$$

New Notion: Digital Frequency!



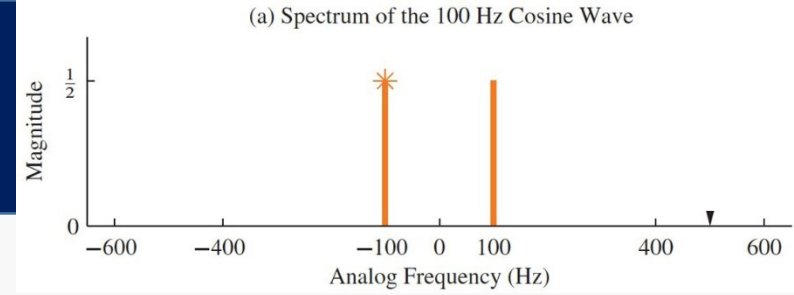
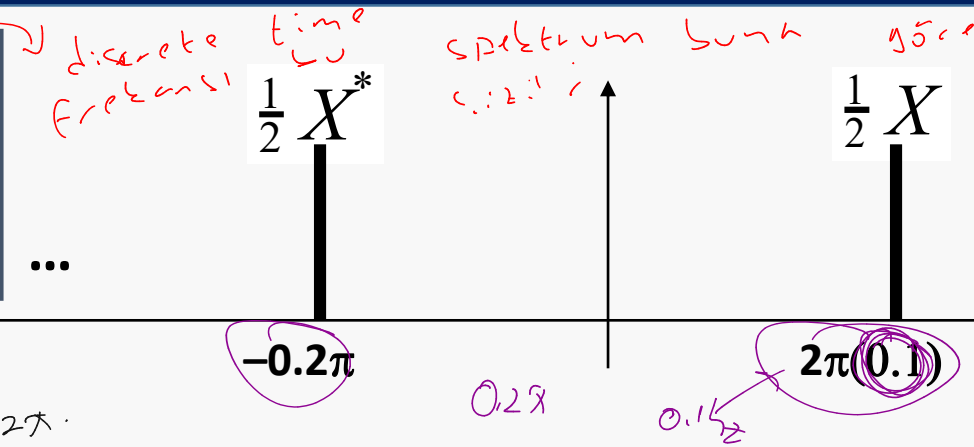
- $\hat{\omega}$ VARIES from **0** to **2π** , as f varies from 0 to the sampling frequency
- UNITS are radians, not rad/sec
 - **DIGITAL FREQUENCY** is NORMALIZED

$$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s}$$
A blue curved arrow originates from the text 'DIGITAL FREQUENCY' in the list above and points to the equation $\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s}$, which is enclosed in a green rectangular box.

Spectrum of Digital Signal

$$\hat{\omega} = 2\pi \frac{f}{f_s}$$

$$f_s = 1 \text{ kHz}$$



$x(t) = A \cos(100 \times 2\pi f t)$
 $f_s = 1000 \text{ Hz}$
 $\hat{\omega} = \frac{2\pi f}{f_s} = \hat{\omega} = \frac{2\pi}{10} = 0.2\pi$
 $x[n] = A \cos(0.2\pi n)$

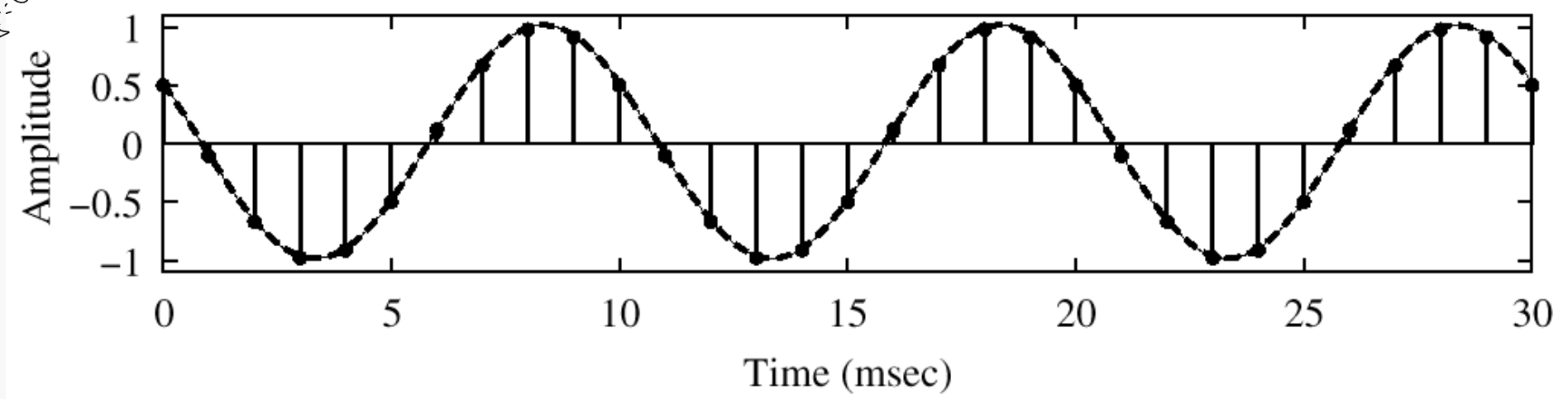
$t = n \cdot T_s = \frac{n}{f_s}$
 $\omega = \frac{2\pi f}{f_s} = \omega \cdot T_s$

$$x[n] = A \cos(2\pi(100)(n/1000) + \varphi)$$

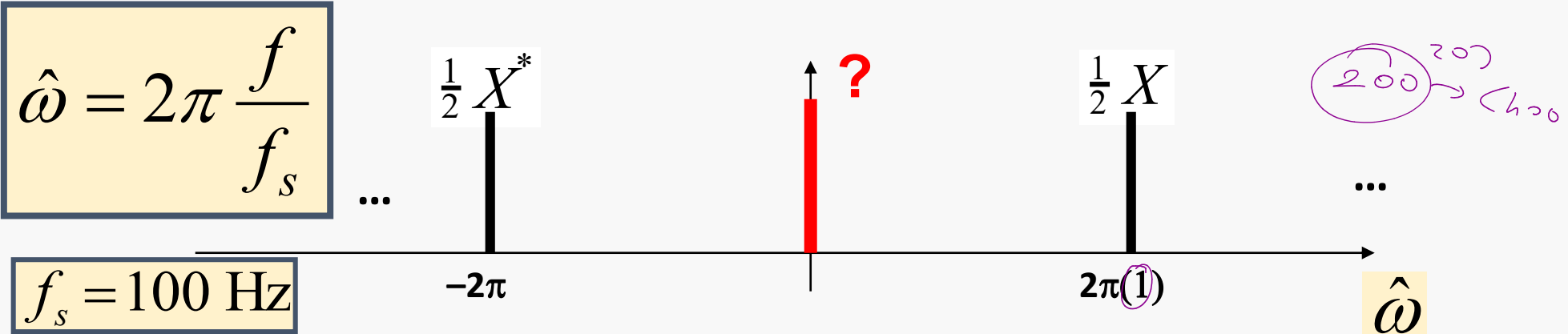
100-Hz Cosine Wave: Sampled with $T_s = 1 \text{ msec}$ (1000 Hz)

$$\omega = \frac{2\pi f}{f_s}$$

$$t = n \cdot T_s = \frac{n}{f_s}$$

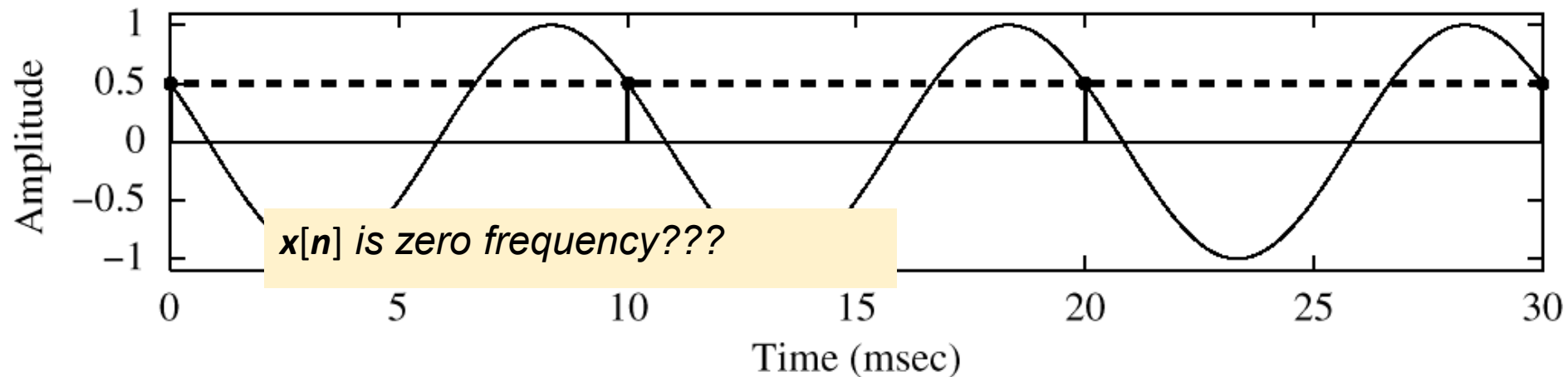


Spectrum of Digital Signal



$$x[n] = A \cos(2\pi(100)(n/100) + \varphi)$$

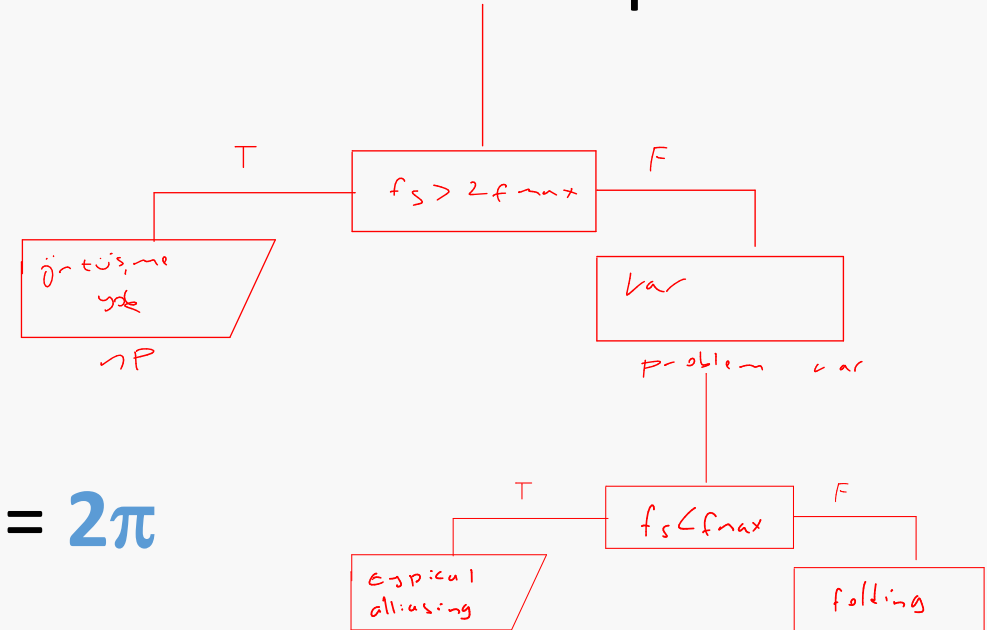
100-Hz Cosine Wave: Sampled with $T_s = 10 \text{ msec}$ (100 Hz)



The Rest of the Story



- Spectrum of $x[n]$ has more than one line for each complex exponential
 - Called ALIASING
 - MANY SPECTRAL LINES
- SPECTRUM is PERIODIC with period = 2π
 - Because



$$A \cos(\hat{\omega}n + \varphi) = A \cos((\hat{\omega} + 2\pi\ell)n + \varphi)$$

Example for clarification



- Other Frequencies give the same

$$x_1(t) = \cos(400\pi t) \quad \text{sampled at } f_s = 1000\text{Hz}$$

$$x_1[n] = \cos(400\pi \frac{n}{1000}) = \cos(0.4\pi n)$$

$$x_2(t) = \cos(2400\pi t) \quad \text{sampled at } f_s = 1000\text{Hz}$$

$$x_2[n] = \cos(2400\pi \frac{n}{1000}) = \cos(2.4\pi n)$$

$$x_2[n] = \cos(2.4\pi n) = \cos(0.4\pi n + 2\pi n) = \cos(0.4\pi n)$$

$$\Rightarrow x_2[n] = x_1[n]$$

$$2400\pi - 400\pi = 2\pi(1000)$$

→ 2 farklı
sinusoidal discrete signal
aynı gibi olur

$$\hat{\omega}$$

$$0.4\pi = \hat{\omega}$$

$$t = n \cdot T_s = \frac{n}{f_s}$$

$$0.4\pi n$$

$$-1.6$$

$$\cos(0.4\pi \cdot 2) = 1$$

$$1200 \leq 2\pi \cdot 3.2$$

GIVEN $x[n]$, we CAN'T KNOW whether it came from a sinusoid at f_0 or $(f_0 + f_s)$ or $(f_0 + 2f_s)$...

Digital Frequency Repeats for each Every 2π



- Other Frequencies give the same

$$\hat{\omega}$$

$$\text{If } x(t) = A \cos(2\pi(f + \ell f_s)t + \varphi)$$

$$t \leftarrow \frac{n}{f_s}$$

$$\text{and we want : } x[n] = A \cos(\hat{\omega}n + \varphi)$$

$$\text{then : } \hat{\omega} = \frac{2\pi(f + \ell f_s)}{f_s} = \frac{2\pi f}{f_s} + \frac{2\pi \ell f_s}{f_s}$$

$$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s} + 2\pi \ell$$

Normalized Frequency

$$\hat{\omega}$$

Normalized Radian Frequency

$$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s} + 2\pi \ell$$

ALIASING

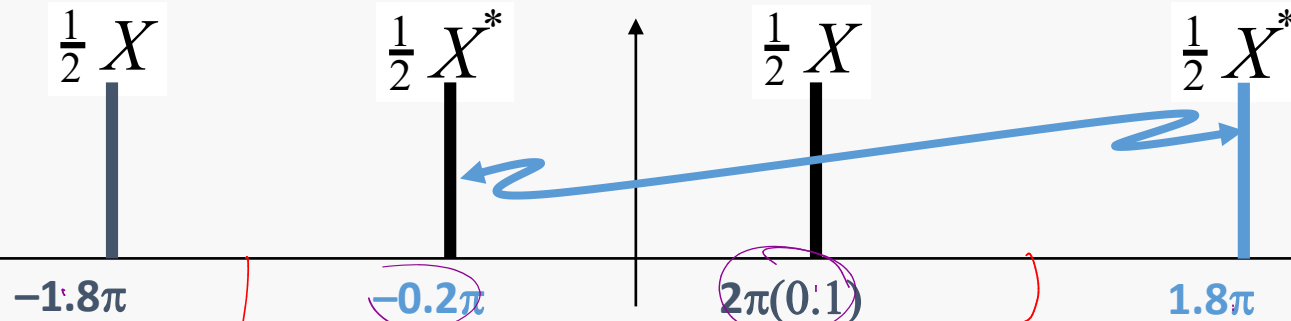
$$\hat{\omega} = \omega T_s = -\frac{2\pi f}{f_s} + 2\pi \ell$$

FOLDED ALIAS

Example Spectrum-1

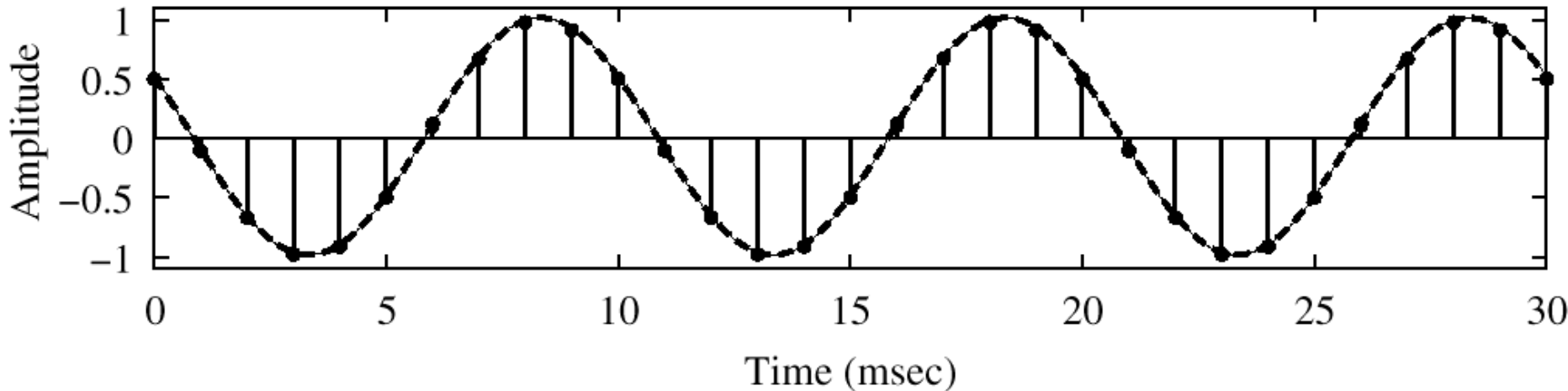
$$\hat{\omega} = 2\pi \frac{f}{f_s}$$

$$f_s = 1 \text{ kHz}$$



$$x[n] = A \cos(2\pi(100)(n/1000) + \varphi)$$

100-Hz Cosine Wave: Sampled with $T_s = 1 \text{ msec}$ (1000 Hz)



Principal alias :

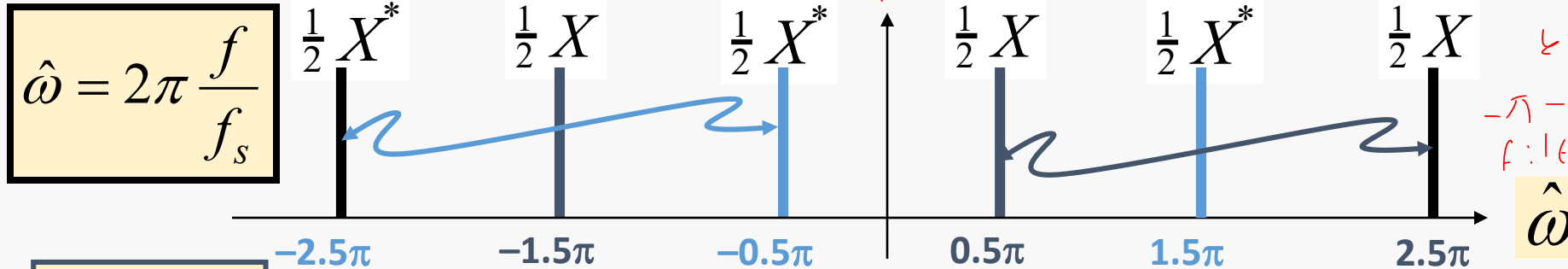
$$f = \frac{\hat{\omega} f_s}{2\pi} = 0.1(1000) = 100 \text{ Hz}$$

$$x(t) = A \cos(2\pi 100t + \varphi)$$

Aliasing Example Spectrum-2



Principal alias is always between $-\pi \leq \hat{\omega} \leq \pi$



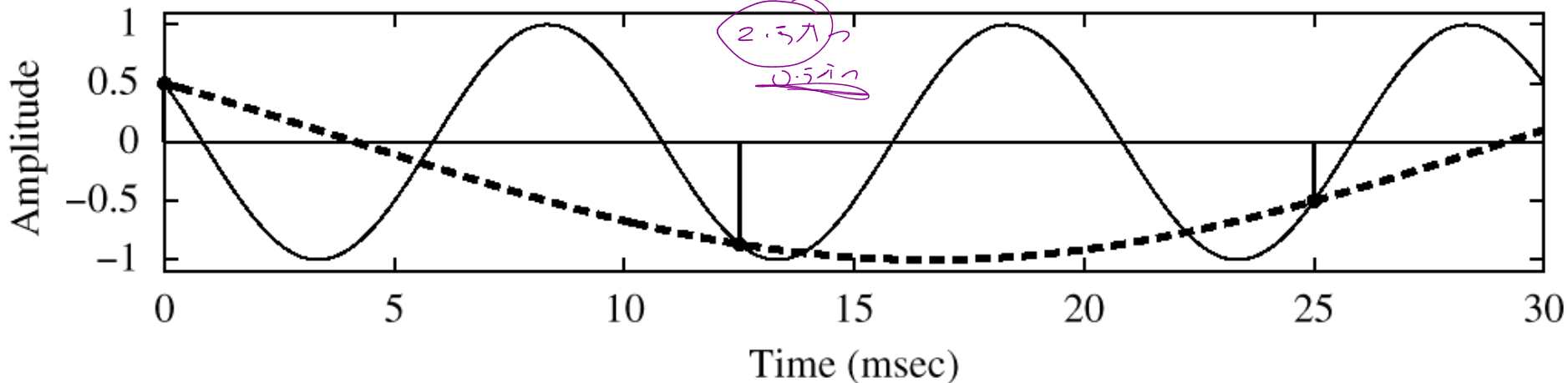
$$\hat{\omega} = 2\pi \frac{f}{f_s}$$

$$f_s = 80 \text{ Hz}$$

$$t = \frac{n}{80}$$

$$x[n] = A \cos(2\pi(100)(n/80) + \varphi)$$

100-Hz Cosine Wave: Sampled with $T_s = 12.5$ msec (80 Hz)



bu sinyal'i geri elde etmek istiyor
kıs: $-\pi - \pi$ aralığını filtreliyoruz
ve dalga'nın f -eşanti
 $f = 0.5 \cdot 80 \rightarrow f = 20 \text{ Hz}$
bulacak
yanlış olamaz

Principal alias :

$$f = \frac{\hat{\omega} f_s}{2\pi} = 0.25(80) = 20 \text{ Hz}$$

$$x(t) = A \cos(2\pi 20t + \varphi)$$

From the book (the same example, more clear)

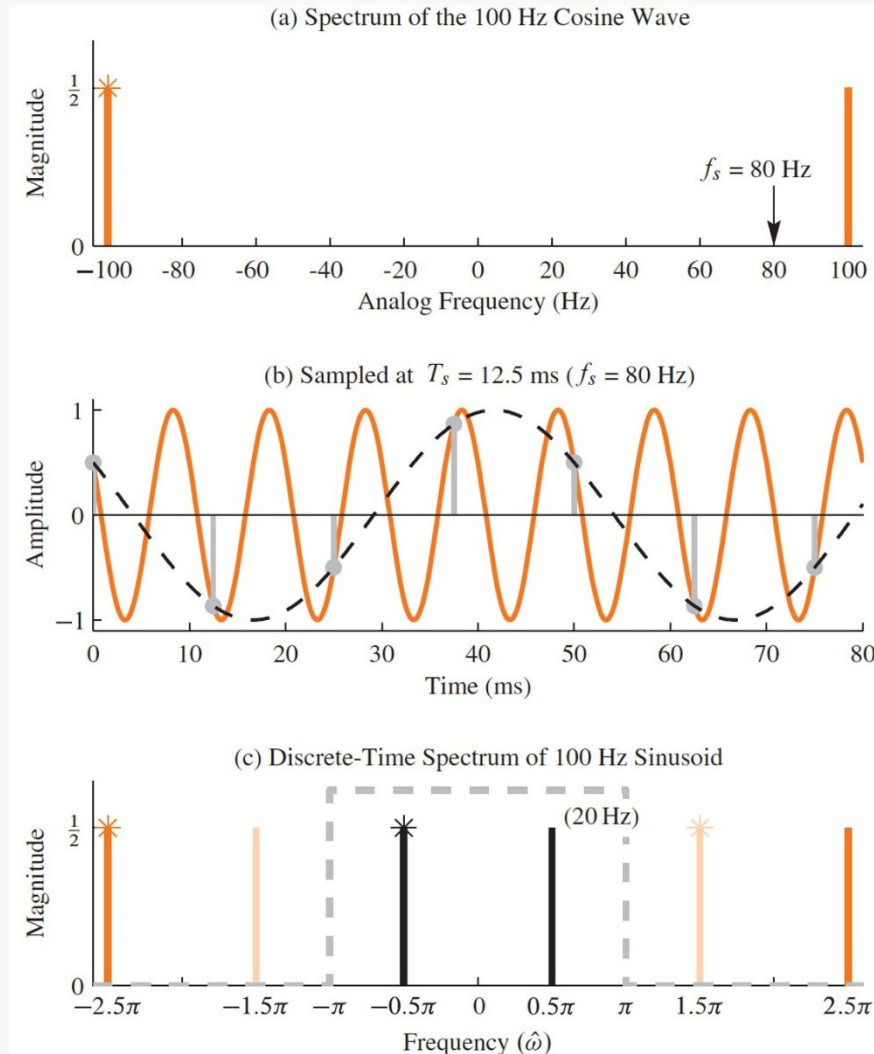


Figure 4-9 Under-sampling a 100 Hz sinusoid at $f_s = 80$ samples/s.

(a) Continuous-time spectrum;

(b) time-domain plot, showing the samples $x[n]$ as gray dots, the original signal $x(t)$ as a continuous **orange** line, and the reconstructed signal $y(t)$ as a dashed black line, which is a 20 Hz sinusoid passing through the same sample points; and

(c) discrete-time spectrum plot, showing the positive and negative frequency components of the original sinusoid at $\pm 2.5\pi$ rad, along with two sets of alias components.

FOLDING (a type of ALIASING)

- EXAMPLE: 3 different $x(t)$; same $x[n]$

$$2\pi \boxed{100}t$$

$$0.1\pi$$

$$\frac{100}{1000}$$

$$f_s = 1000$$

$$\cos(2\pi(100)t) \rightarrow \cos[2\pi(0.1)n]$$

$$\cos(2\pi(1100)t) \rightarrow \cos[2\pi(1.1)n] = \cos[2\pi(0.1)n]$$

$$\cos(2\pi(900)t) \rightarrow \cos[2\pi(0.9)n]$$

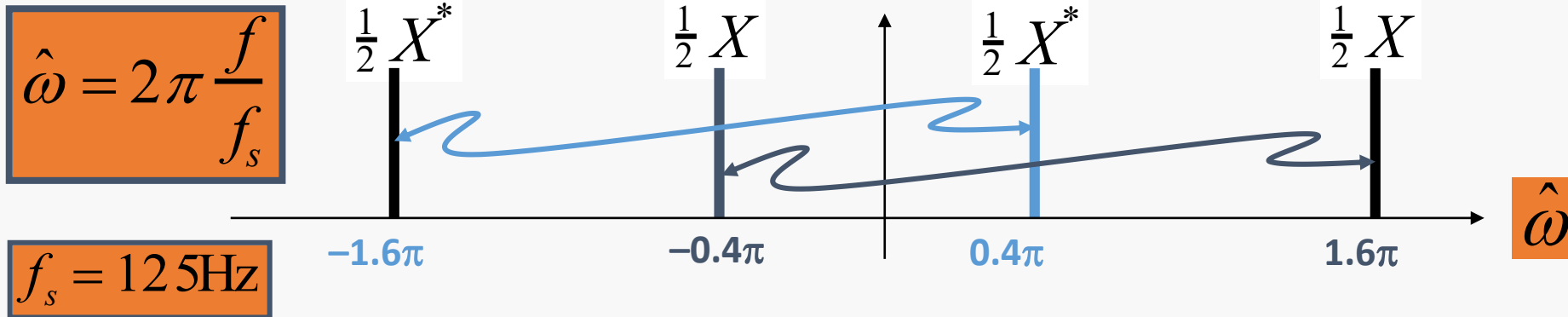
$$= \cos[2\pi(0.9)n - 2\pi n] = \cos[2\pi(-0.1)n] = \cos[2\pi(0.1)n]$$

$$\hat{\omega} = 2\pi \frac{100}{1000} = 2\pi(0.1)$$

- 900 Hz “folds” to 100 Hz when $f_s = 1\text{kHz}$

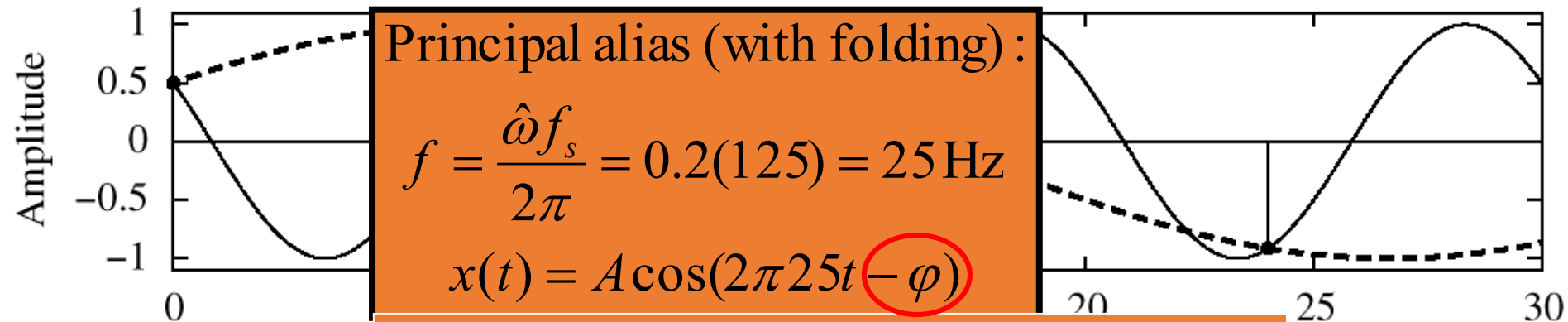
Example Folding Case

fold $\leftarrow f_s < f_{max}$



$$x[n] = A \cos(2\pi(100)(n/125) + \varphi)$$

100-Hz Cosine Wave: Sampled with $T_s = 8$ msec (125 Hz)



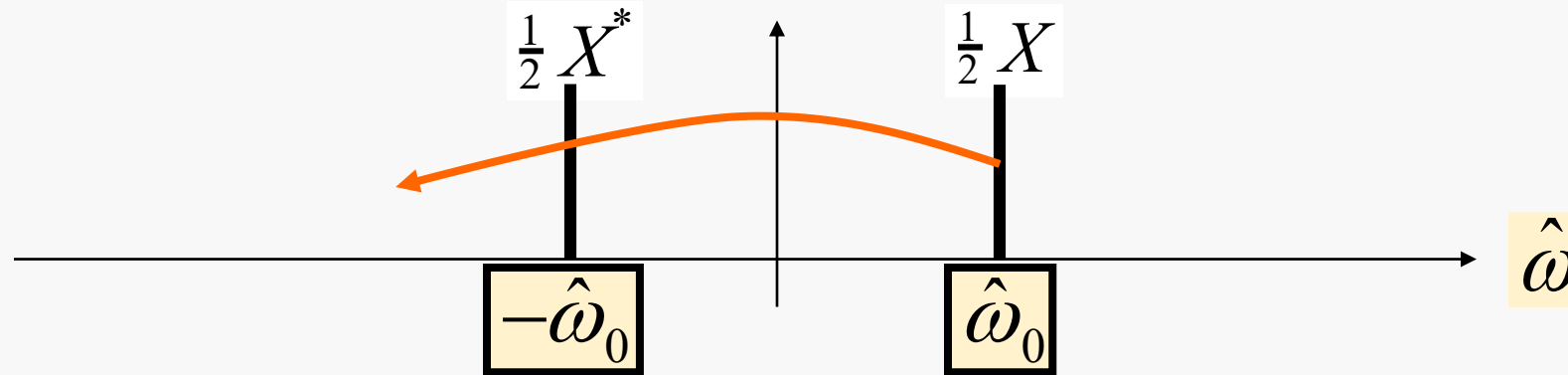
Principal alias is always between $-\pi \leq \hat{\omega} \leq \pi$

Aug 2016

SPECTRUM Explanation of SAMPLING THEOREM



- How do we prevent aliasing?
- Guarantee original signal is principal alias:



$$\hat{\omega}_0 - 2\pi < -\hat{\omega}_0 \Rightarrow \hat{\omega}_0 < \pi$$
$$\hat{\omega}_0 = \frac{2\pi f_0}{f_s} < \pi \Rightarrow f_0 < \frac{f_s}{2}$$

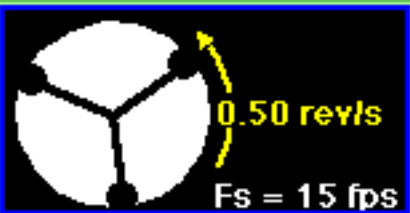
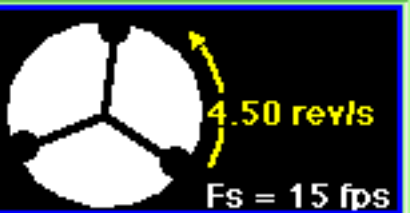
Be Careful:

<https://www.youtube.com/watch?v=qgvuQGY946g>



Strobe Demo



CONSTANT DISK SPEED: NO ALIASING	CONSTANT SPEED: FOLDING	CONSTANT SPEED: ALIASING
 0.75 rev/s Fs = 15 fps	 12.50 rev/s Fs = 15 fps	 15.50 rev/s Fs = 15 fps
 1.50 rev/s Fs = 15 fps	 14.00 rev/s Fs = 15 fps	 16.00 rev/s Fs = 15 fps
 3.00 rev/s Fs = 15 fps	 14.50 rev/s Fs = 15 fps	 31.00 rev/s Fs = 15 fps
 0.50 rev/s Fs = 15 fps	 4.50 rev/s Fs = 15 fps	 5.50 rev/s Fs = 15 fps

$$1 \quad 14 \quad 0.75 \times 2$$

$$14.25$$

<https://dspfirst.gatech.edu/chapters/04samplin/demos/strobe/index.html>

$$11.8 \times 10^{18} \quad 0.1 \times$$

<https://dspfirst.gatech.edu/chapters/04samplin/demos/synstrob/index.html>

$$2 \times 10^{-5}$$

$$15$$

Digital to Analog Reconstruction



- Create continuous $y(t)$ from $y[n]$

- IDEAL D-to-A:

- If you have formula for $y[n]$
- Invert sampling ($t=nT_s$) by $\mathbf{n=f_s t}$
- $y[n] = A\cos(0.2\pi n + \phi)$ with $f_s = 8000$ Hz
- $y(t) = A\cos(0.2\pi(8000t) + \phi) = A\cos(2\pi(\mathbf{800})t + \phi)$

$n \cdot T_s \leq t \leq f_s$
 $T_s = \frac{1}{f_s}$

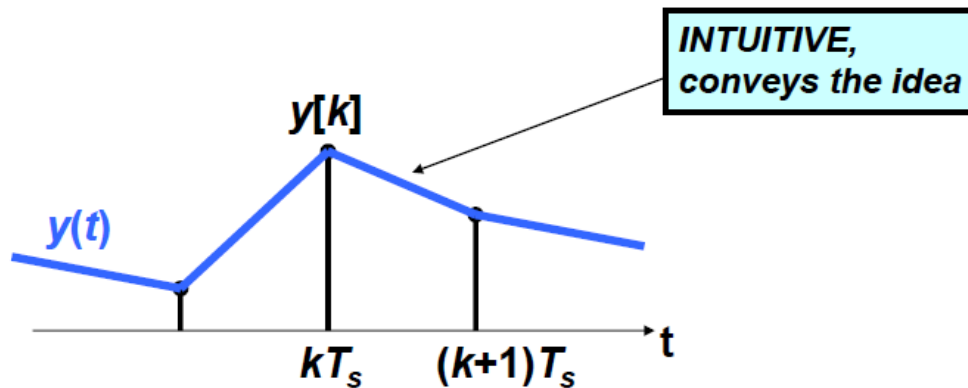
$t \neq f_s = ?$

1600

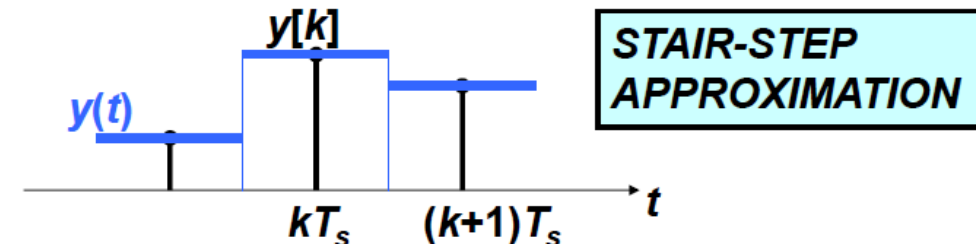
- RECONSTRUCT THE SMOOTHEST ONE
 - THE LOWEST FREQ, if $y[n] = \text{sinusoid}$

Ayrıca analoge çevirmenin
2 yolu var
1-) Basamak yaklaşımı
2-) Enterpolasyon

- CONVERT STREAM of NUMBERS to $x(t)$
- “CONNECT THE DOTS”
- INTERPOLATION

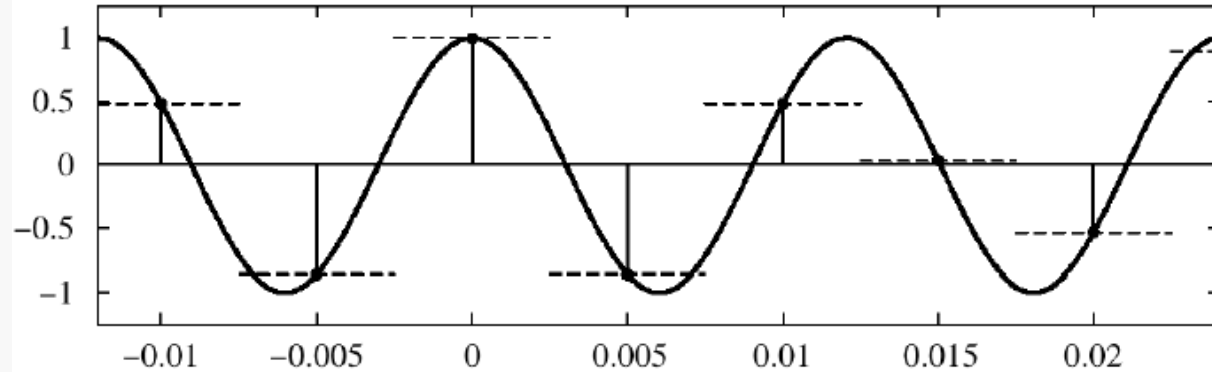


- CONVERT $y[n]$ to $y(t)$
 - $y[k]$ should be the value of $y(t)$ at $t = kT_s$
 - Make $y(t)$ equal to $y[k]$ for
 - $kT_s - 0.5T_s < t < kT_s + 0.5T_s$

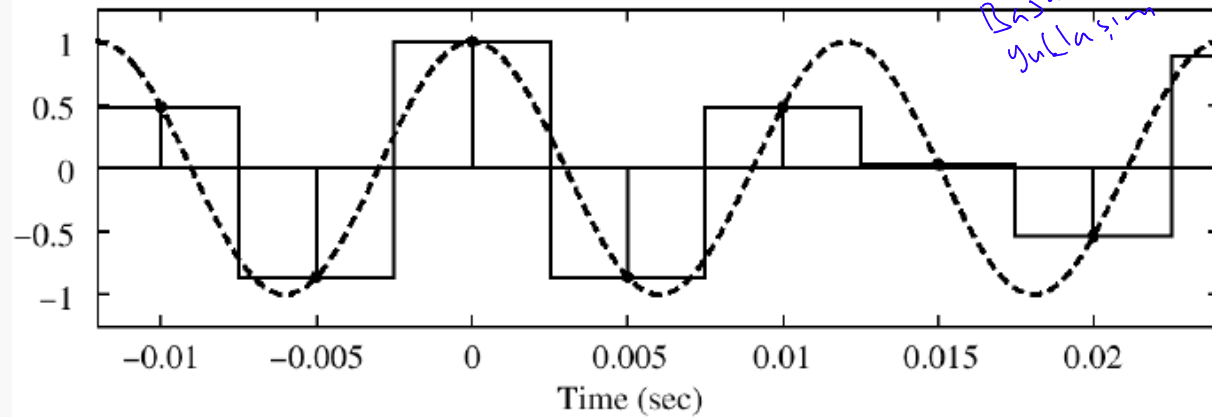


STAIR-STEP APPROXIMATION

Sampling and Zero-Order Reconstruction: $f_0 = 83$ $f_s = 200$

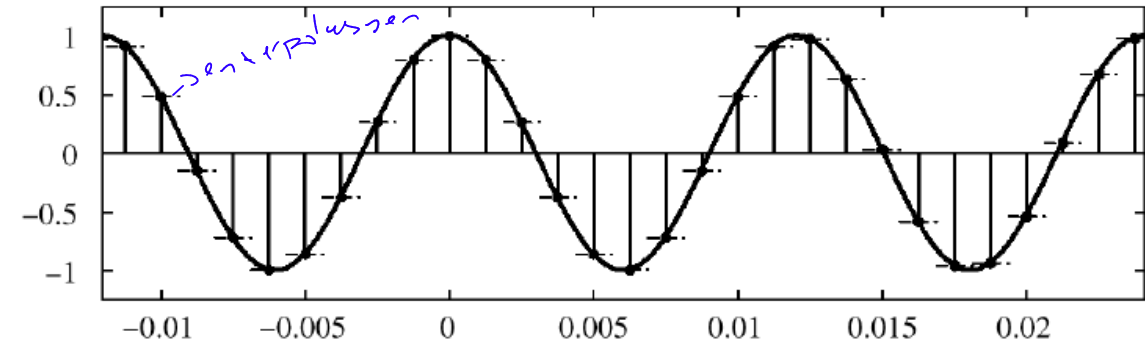


Original and Reconstructed Waveforms



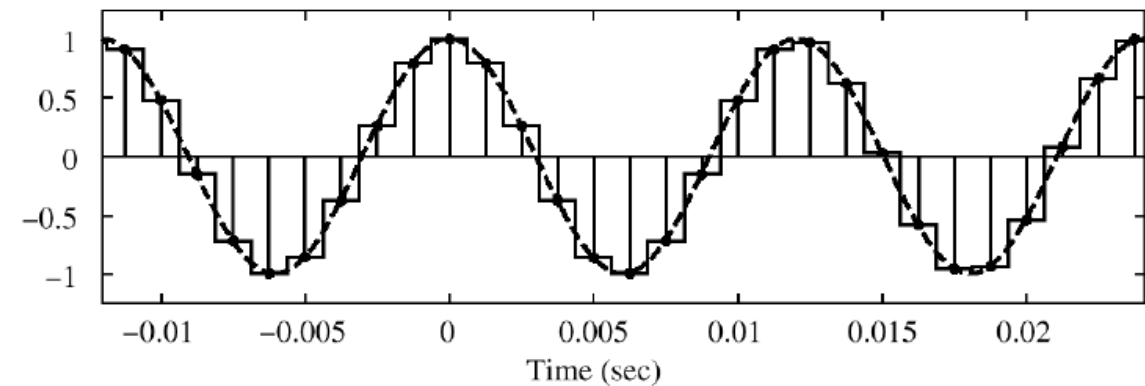
OVER-SAMPLING CASE

Sampling and Zero-Order Reconstruction: $f_0 = 83$ $f_s = 800$

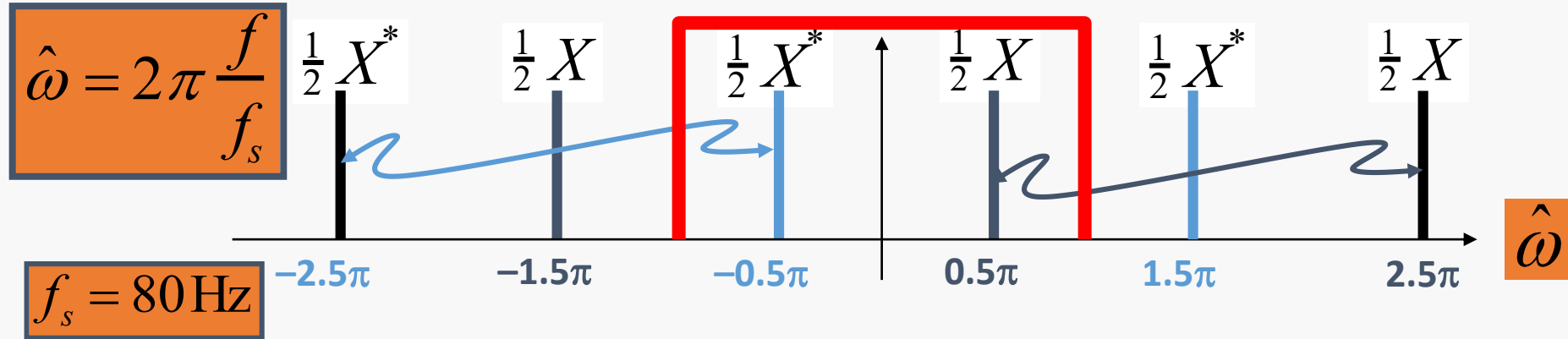


EASIER TO RECONSTRUCT

Original and Reconstructed Waveforms

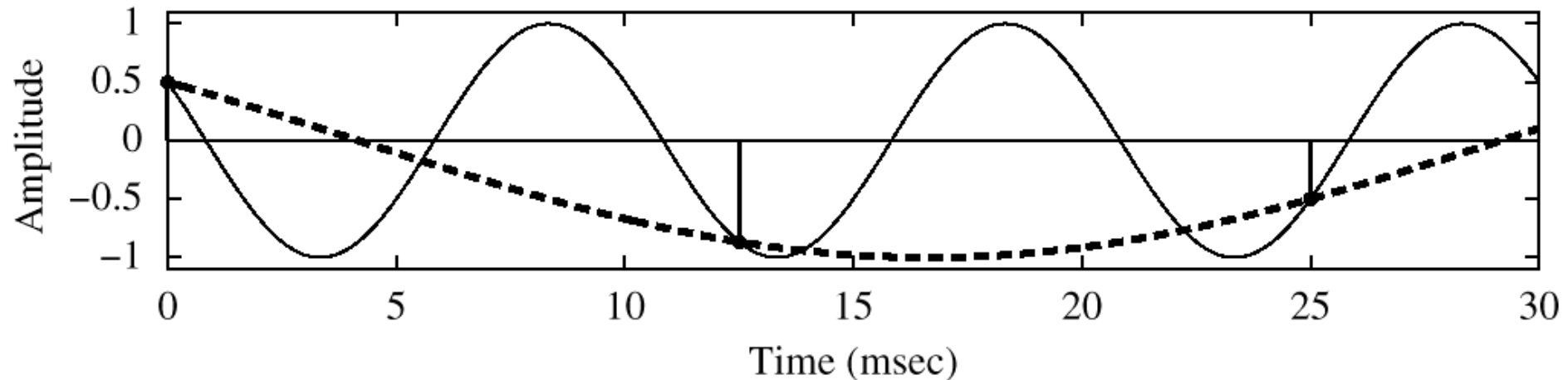


SPECTRUM (ALIASING CASE)

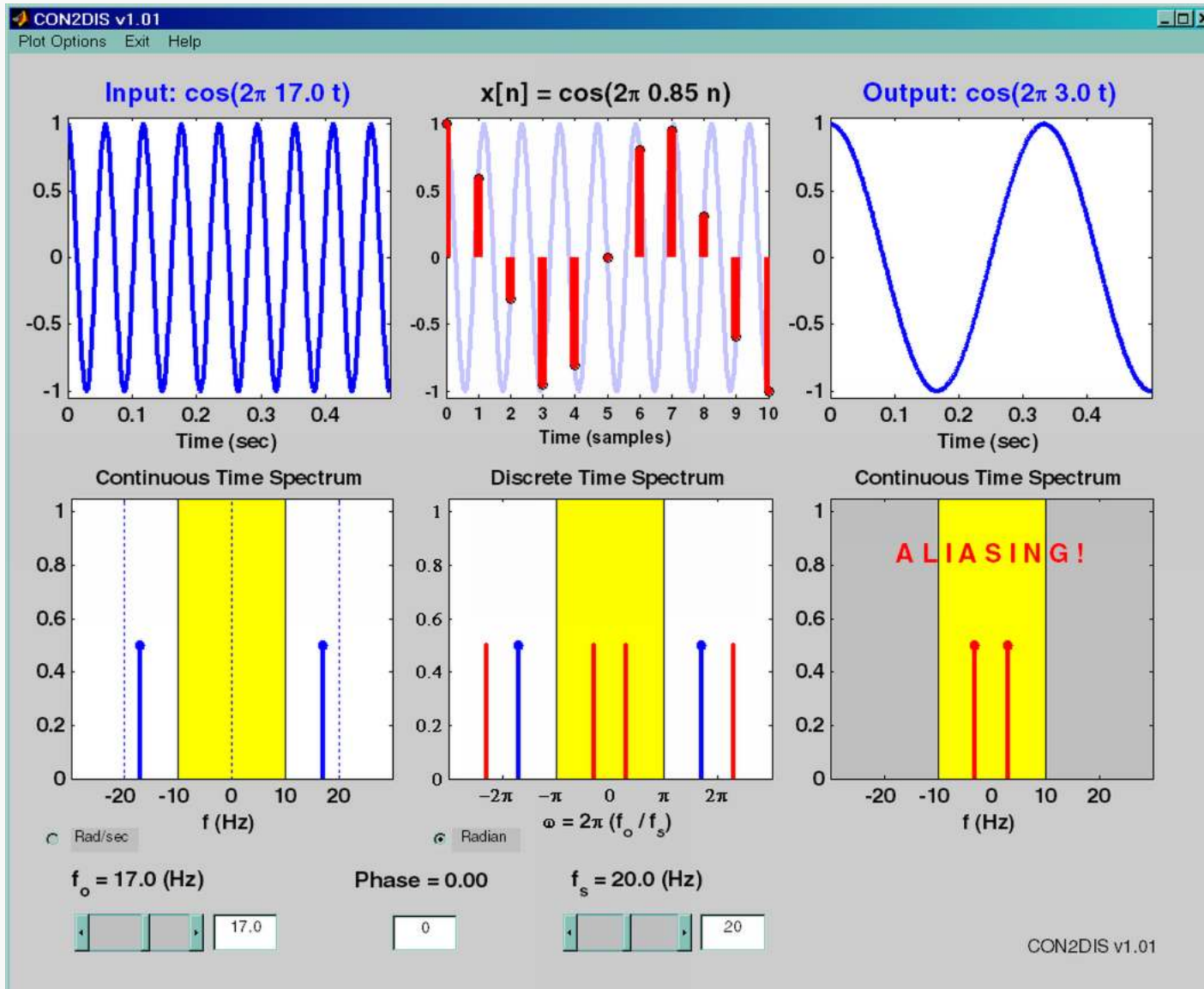


$$x[n] = A \cos(2\pi(100)(n/80) + \varphi)$$

100-Hz Cosine Wave: Sampled with $T_s = 12.5$ msec (80 Hz)

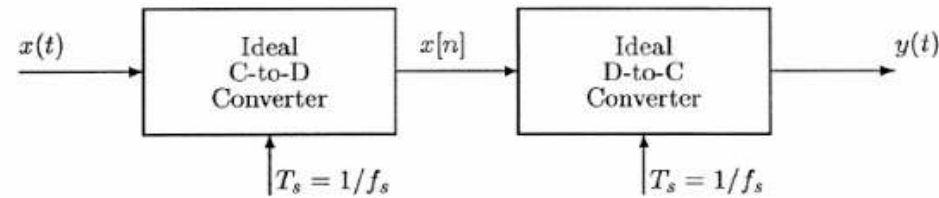


Let's Analyze with MATLAB



<https://dspfirst.gatech.edu/matlab/#con2dis>

Example - 1



Suppose that the output of the D-to-C converter in the system above is found to be

$$y(t) = 2 + 10 \cos(2\pi(150)t + \pi/3)$$

$$\frac{150}{400}$$

$$\frac{3}{8} \pi$$

when the sampling rate is $f_s = 1/T_s = 400$ samples/second.

- (a) Give an equation for $x[n]$ in terms of cosine functions. Write your answer on the line below.

With no aliasing, going through a D/C converter is the inverse of going through the C/D. Therefore, we can get $x[n]$ by passing $y(t)$ through a C/D converter.

$$\begin{aligned} \text{Answer: } x[n] &= 2 + 10 \cos(2\pi(150)n/400 + \pi/3) \\ &= 2 + 10 \cos(3n\pi/4 + \pi/3) \end{aligned}$$

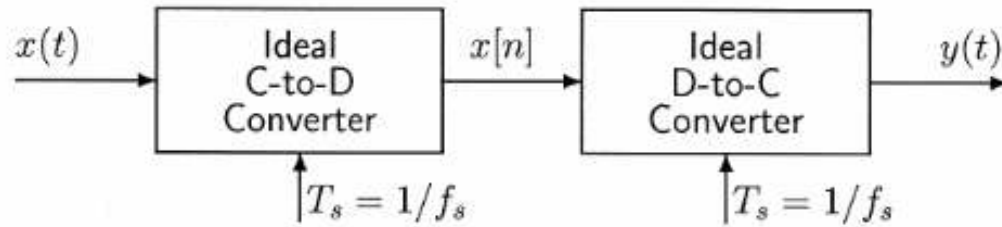
- (b) Determine two *different* input signals $x(t) = x_1(t)$ and $x(t) = x_2(t)$ that could have produced the given output of the D-to-C converter. All of the frequencies in your answers must be positive and less than 400 Hz. Write your answers for both inputs on the lines below.

$$\text{Answer: } x_1(t) = \frac{2 + 10 \cos(2\pi(150)t + \pi/3)}{(\text{no aliasing})}$$

$$\text{Answer: } x_2(t) = 2 + 10 \cos(2\pi(250)t + -\pi/3)$$

We have folding in the second case.

Example-2



Suppose that the continuous-time input $x(t)$ to the above system is given as

$$x(t) = \cos(14000\pi t) + \cos(2000\pi t) + \cos(1000\pi t).$$

$$f_{\max} = 7,000 \text{ Hz}$$

(a) What sampling rate is required such that no aliasing occurs for $x(t)$?

$$f_s = 14,000 \text{ Hz}$$

(c) Given that $x(t) = \cos(25000\pi t)$ and $f_s = 10000$ samples/second, write a simplified expression for the output $y(t)$ in terms of cosine functions.

$$y(t) = \cos(5000\pi t)$$

NOTE THAT ALIASING OCCURS.

$$x[n] = \cos\left(\frac{25000\pi n}{10000}\right) = \cos(0.5\pi n)$$

$$y(t) = \cos(0.5\pi t 10000) = \cos(5000\pi t)$$

$$0.5 \cdot 10000$$

$$5000\pi t$$

Example-3

PROBLEM:

The “spectrum” diagram gives the frequency content of a signal.

- (a) Draw a sketch of the spectrum of $x(t)$ which is “cosine-times-sine”

$$x(t) = \cos(50\pi t) \sin(700\pi t)$$

Label the frequencies and complex amplitudes of each component.

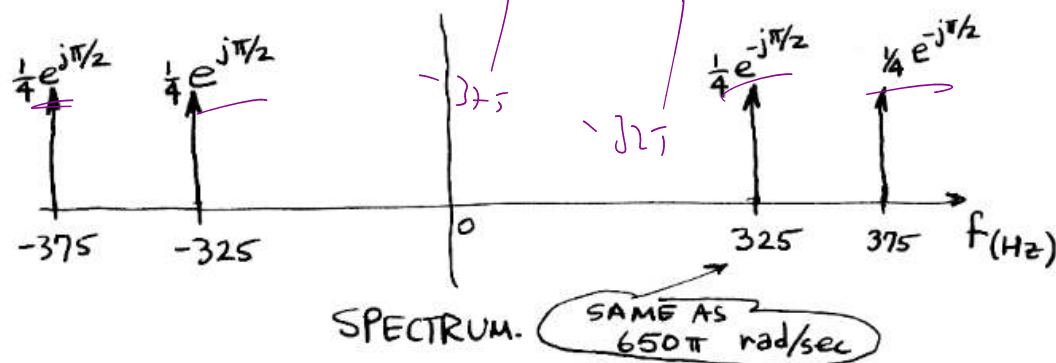
- (b) Determine the minimum sampling rate that can be used to sample $x(t)$ without any aliasing.

$$x(t) = \cos(50\pi t) \sin(700\pi t)$$

$$(a) \quad x(t) = \left(\frac{1}{2} e^{j50\pi t} + \frac{1}{2} e^{-j50\pi t} \right) \left(\frac{1}{2j} e^{j700\pi t} - \frac{1}{2j} e^{-j700\pi t} \right)$$

$$= \frac{1}{4j} e^{j750\pi t} + \frac{1}{4j} e^{j650\pi t} - \frac{1}{4j} e^{-j650\pi t} - \frac{1}{4j} e^{-j750\pi t}$$

SAME AS $\frac{1}{4} e^{j\pi/2}$ SAME AS $\frac{1}{4} e^{+j\pi/2}$



$$\frac{1}{2} \cdot (e^{50\pi t} + e^{-50\pi t}) \cdot \frac{1}{2j} \cdot (e^{700\pi t} - e^{-700\pi t})$$

- (b) Sampling Thm says sample at a rate greater than **two times** the highest freq.

$$\text{HIGHEST FREQ} = 375 \text{ Hz}$$

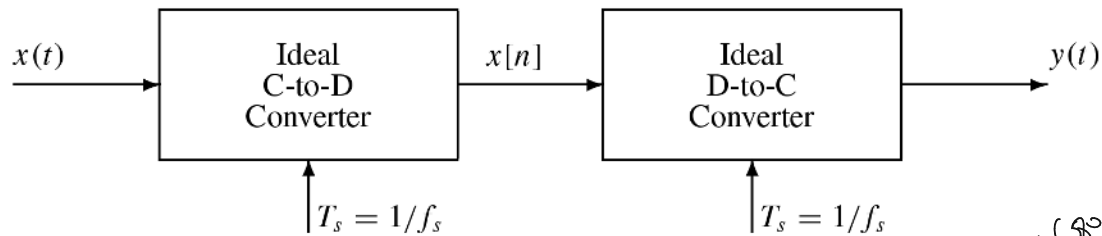
$$\Rightarrow f_s \geq 750 \text{ Hz.}$$

Example-4



PROBLEM:

Consider the following system.



Suppose that the output of the C-to-D converter is

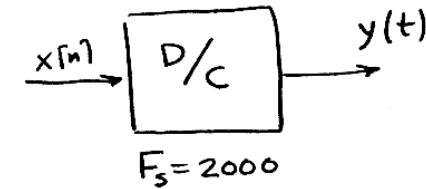
$$x[n] = 5 + 8 \cos(0.4\pi n) + 4 \cos(0.8\pi n + \pi/3)$$

when the sampling rate is $f_s = 1/T_s = 2000$ samples/second. Determine the output $y(t)$ of the ideal D-to-C converter.

$$y(t) = 5 + 8 \cos(800\pi t) + 4 \cos(1600\pi t + \pi/3)$$

$$5 + 8 \cos(800\pi t) + 4 \cos(1600\pi t + \pi/3)$$

$$x[n] = 5 + 8 \cos(0.4\pi n) + 4 \cos(0.8\pi n + \pi/3)$$



For discrete to continuous, we replace "n" with $F_s t$

$$y(t) = x[n] \Big|_{n=F_s t}$$

$$= 5 + 8 \cos(0.4\pi(2000)t) + 4 \cos(0.8\pi(2000)t + \pi/3)$$

$$= 5 + 8 \cos(2\pi(400)t) + 4 \cos(2\pi(800)t + \pi/3)$$